

ASSIGNMENT

- 1. Explain the architecture of SIC and SIC/XE with important features.**
- 2. How is forward reference handled in One pass Assembler?**
- 3. Explain machine dependent assembler features.**
- 4. What is Absolute Loader. Explain with algorithm**
- 5. Explain major difference between System Software and Application Software.**
- 6. Define Loader. Differentiate between Linking Loader from Linkage Editor.**
- 7. Write about Instruction format, addressing mode and registers of SIC/XE architecture.**
- 8. Discuss the role of data structures OPTAB, SYMTAB, LOCCTR.**
- 9. What is relocation? How is relocation Carried out in Loader.**
- 10. Write a SIC program to find sum of two numbers and save result in Sum Variable.**

UNIT 3

- 11. What are Loader design options? Briefly explain two of them.**
- 12. What are the basic functions of a loader? Explain the design of an absolute loader with a suitable algorithm and a simple example.**
- 13. Explain the machine-dependent features of a loader, focusing on Relocation and Program Linking.**
- 14. What are the machine-independent features of a loader? Explain Automatic Library Search and various Loader Options in detail.**
- 15. Differentiate between an absolute loader, a relocating loader, and a linking loader.**

UNIT 4

- 16. Define a Macro Processor. Explain the basic functions of a macro processor, including Macro Definition and Macro Expansion, with a simple example.**
- 17. What are the data structures used in the design of a macro processor? Explain each briefly**
- 18. What are the machine-independent features of a macro processor? Discuss Conditional Macro Expansion and the Generation of Unique Labels with suitable examples.**
- 19. What is Recursive Macro Expansion? How does a macro processor handle recursive macro definitions? Explain with a practical example and a diagram.**
- 20. Explain how a macro processor generates unique labels during macro expansion. Why is this necessary? Provide an example.**

UNIT 6

- 21. Explain the Booch Macro and Micro processes in the context of Object-Oriented Design of an assembler**
- 22. Describe the Object Diagram for an assembler. Identify the key classes, their attributes, and their relationships. Draw a neat diagram to support your answer.**
- 23. Explain the Interaction Diagram for an assembler. How does it show the flow of messages between objects during the assembly process?**
- 24. What is Object-Oriented Programming (OOP)? Explain the key principles of OOP and how they are applied in the design of system software like an assembler**

UNIT 5

- 25. What is a Compiler? Explain the fundamental phases of a compiler with a neat block diagram**
- 26. Describe the role of the Lexical Analyzer and Syntactic Analyzer (Parser) in the compilation process.**
- 27. Describe Recursive Descent Parsing. How does it differ from Operator Precedence Parsing?**
- 28. Discuss the concept of Intermediate Code Generation. Why is it an important phase in a compiler?**